

Course: Linear Algebra

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Basis in vector space

A set of vectors $\vec{V} = \{v_1, v_2, v_3, \dots, v_n\}$ in a vector space is said to have basis for v if it satisfies two conditions:

1. Set of vectors $\vec{V} = \{v_1, v_2, v_3, \dots, v_n\}$ must be span.
2. Set of vectors $\vec{V} = \{v_1, v_2, v_3, \dots, v_n\}$ must be linearly independent.

Example:1. Show that the following set is a basis for R^3
 $V = \{(1,0,0), (0,1,0), (0,0,1)\}$.

Solution,

If $V = \{(1,0,0), (0,1,0), (0,0,1)\}$ is the basis for R^3 then it satisfies two conditions:

Condition (i). Set of vectors $\vec{V} = \{v_1, v_2, v_3, \dots, v_n\}$ must be span. Hence,

Apply the condition for spanning

$$u = (c_1 v_1 + c_2 v_2 + c_3 v_3 + \dots + c_n v_n)$$

Or

$$u_1, u_2, u_3 = c_1 v_1 + c_2 v_2 + c_3 v_3 \quad (1)$$

By equating corresponding components, we get the system of linear equations

$$u_1, u_2, u_3 = c_1(1,0,0) + c_2(0,1,0) + c_3(0,0,1)$$

$$\square |A| = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} \neq 0$$

Since $|A| \neq 0$. So, Set of vectors can be expressed for linear combination. Hence, set of vectors is spanning in R^3 .

Condition (ii). Set of vectors $\vec{V} = \{v_1, v_2, v_3, \dots, v_n\}$ must be linearly independent. Hence, applying the formula of Linear independence

$$c_1 v_1 + c_2 v_2 + c_3 v_3 + c_4 v_4 + \dots + c_n v_n = 0$$

$$c_1(1,0,0) + c_2(0,1,0) + c_3(0,0,1) = (0,0,0)$$

Solving this , we get trivial solution as $c_1 = 0$, $c_2 = 0$ and $c_3 = 0$. so the set of vectors is linearly independent.

As both conditions are satisfied hence set of vectors is the basis for R^3 .

Example:1. Show that the following set is a basis for R^3
 $V = \{(1,2,3), (0,1,2), (-2,0,1)\}$.

Solution,

If $V = \{(1,2,3), (0,1,2), (-2,0,1)\}$ is the basis for R^3 then it satisfies two conditions:

Condition (i). Set of vectors $\vec{V} = \{v_1, v_2, v_3, \dots, v_n\}$ must be span. Hence,

Apply the condition for spanning

$$u = (c_1 v_1 + c_2 v_2 + c_3 v_3 + \dots + c_n v_n)$$

Or

$$u_1, u_2, u_3 = c_1 v_1 + c_2 v_2 + c_3 v_3 \quad (1)$$

By equating corresponding components, we get the system of linear equations

$$u_1, u_2, u_3 = c_1(1, 2, 3) + c_2(0, 1, 2) + c_3(-2, 0, 1)$$

$$\begin{array}{lcl} c_1 - 2c_3 = u_1 \\ 2c_1 + c_2 = u_2 \\ 3c_1 + 2c_2 + c_3 = u_3 \end{array} \quad \square \quad |A| = \begin{vmatrix} 1 & 0 & -2 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{vmatrix} \neq 0$$

Since $|A| \neq 0$. So, Set of vectors can be expressed for linear combination. Hence, set of vectors is spanning in R^3 .

Condition (ii). Set of vectors $\vec{V} = \{v_1, v_2, v_3, \dots, v_n\}$ must be linearly independent. Hence, applying the formula of Linear independence

$$c_1 v_1 + c_2 v_2 + c_3 v_3 + c_4 v_4 + \dots + c_n v_n = 0$$

$$c_1(1,2,3) + c_2(0,1,2) + c_3(-2,0,1) = (0,0,0)$$

$$c_1 - 2c_3 = 0, \quad 2c_1 + c_2 = 0, \quad 3c_1 + 2c_2 + c_3 = 0$$

Since $|A| \neq 0$ this means that it has trivial solution, hence the set of vectors is linearly independent.

As both conditions are satisfied hence set of vectors is the basis for R^3 .

Assignment Work

Example:1. Show that the following set is a basis for R^3
 $V = \{(-1, 2, 5), (3, 0, 3), (5, 1, 8)\}.$

Nullity

A matrix that satisfies the condition

$$\text{Nullity of } (\mathbf{matrix}) = (\text{Number of unknown}) - \text{Rank}(\mathbf{matrix})$$

Example: Find the nullity of matrix

$$A = \begin{bmatrix} 1 & 0 & -2 & 1 & 0 \\ 0 & -1 & -3 & 1 & 3 \\ -2 & -1 & 1 & -1 & 3 \\ 0 & 3 & 9 & 0 & -12 \end{bmatrix}$$

Solution, First of all, we find rank of above matrix which is

$$A = \begin{bmatrix} 1 & 0 & -2 & 0 & 1 \\ 0 & 1 & 3 & 0 & -4 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \text{Rank} = 3$$

$$\text{Nullity of } (A) = (n) - \text{Rank}(A) = 5 - 3 = 2$$

Linear Dependence And Independence of Functions By Wronskian Matrix

Unfortunately there is no general method that can be used to determine whether a set of functions is linearly dependent or independent. However there does exist a matrix for establishing linear dependence or independence in the circumstances of functions. The matrix is known as Wronskian matrix which is used as a tool for deciding whether the functions are linear dependent or independent.

Statement: Let $f_1(x), f_2(x), f_3(x), \dots, f_n(x)$ are the functions then linear dependence or independence is checked by Wronskian matrix below:

$$W(x) = \begin{vmatrix} f_1(x) & f_2(x) & \dots & f_n(x) \\ f_1'(x) & f_2'(x) & \dots & f_n'(x) \\ f_1''(x) & f_2''(x) & \dots & f_n''(x) \\ f_1^{(n-1)}(x) & f_2^{(n-1)}(x) & \dots & f_n^{(n-1)}(x) \end{vmatrix}$$

Procedure

- (i). If Wronskian matrix $W(x) = 0$ then set of functions is linearly dependent.
- (ii). If Wronskian matrix $W(x) \neq 0$ then set of functions is linearly independent.

Example:1. If $f_1(x) = 1, f_2(x) = e^x$ and $f_3(x) = e^{2x}$ be three functions. Check whether the functions are linearly dependent or independent using Wronskian matrix method.

Solution, Given the functions with derivatives,

$$f_1(x) = 1, f_2(x) = e^x \text{ and } f_3(x) = e^{2x} \text{ then}$$

$$f_1^*(x) = 0, f_2^*(x) = e^x \text{ and } f_3^*(x) = 2e^{2x}$$

$$f_1^{**}(x) = 0, f_2^{**}(x) = e^x \text{ and } f_3^{**}(x) = 4e^{2x}$$

Now applying Wronskian matrix

$$W(x) = \begin{vmatrix} f_1(x) & f_2(x) & \dots & f_n(x) \\ f_1^*(x) & f_2^*(x) & \dots & f_n^*(x) \\ f_1^{**}(x) & f_2^{**}(x) & \dots & f_n^{**}(x) \\ f_1^{(n-1)}(x) & f_2^{(n-1)}(x) & \dots & f_n^{(n-1)}(x) \end{vmatrix}$$

Now formulate Wronskian matrix according to given functions

$$W(x) = \begin{vmatrix} f_1(x) & f_2(x) & f_3(x) \\ f_1'(x) & f_2'(x) & f_3'(x) \\ f_1''(x) & f_2''(x) & f_3''(x) \end{vmatrix}$$

Putting values in above matrix,

$$W(x) = \begin{vmatrix} 1 & e^x & e^{2x} \\ 0 & e^x & 2e^{2x} \\ 0 & e^x & 4e^{2x} \end{vmatrix} = 2e^{3x}$$

Since, Wronskian matrix $W(x) \neq 0$ then set of functions is linearly independent.

Assignment Work

Test the Wronskian matrix $W(x)$ for whether functions are linearly dependent or independent given below:

1. $f_1(x) = 1, f_2(x) = x$ and $f_3(x) = e^x$
2. $f_1(x) = 1, f_2(x) = x$ and $f_3(x) = x^2$
3. $f_1(x) = e^x, f_2(x) = xe^x$ and $f_3(x) = x^2e^{2x}$
4. $f_1(x) = \sin x, f_2(x) = \cos x$ and $f_3(x) = x \cos x$
5. $f_1(x) = (x - 1)(x + 2),$
 $f_2(x) = x(x + 2)$ and
 $f_3(x) = x(x - 1).$